higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE N4
NQF LEVEL 4

27 FEBRUARY 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 1.1.1 Form factor is the ratio of the RMS value to the average value.

1.1.2 Peak factor shows the ratio of the peak or maximum value to the RMS value.

(2 × 2) (4)

1.2 1.2.1

$$R_s = R_2 + R_3 + R_4$$

$$= 12 + 5 + 3$$

$$= 20\ \Omega$$

$$R_P = \frac{R_s R_5}{R_s + R_5}$$

$$= \frac{20 \times 30}{20 + 30}$$

$$= 12\ \Omega$$

$$R_T = R_1 + R_P$$

$$= 2 + 12$$

$$= 14\ \Omega$$

ALTERNATIVE:

$$R_s = R_2 + R_3 + R_4$$

$$= 12 + 5 + 3$$

$$= 20\ \Omega$$

$$\frac{1}{Rp} = \frac{1}{R_S} + \frac{1}{R_5}$$

$$\frac{1}{Rp} = \frac{1}{20} + \frac{1}{30}$$

$$\frac{1}{Rp} = \frac{3+2}{60}$$

$$Rp = \frac{60}{5} = 12\ \Omega$$

$$R_T = R_1 + R_P$$

$$= 2 + 12$$

$$= 14\ \Omega$$

(5)

1.2.2

$$I_T = \frac{V_T}{R_T}$$

$$= \frac{10}{14}$$

$$= 0{,}714\ A$$

(2)

1.2.3

$$I_{R5} = \frac{R_S}{R_5 + R_S} I_T$$

$$= \frac{20}{30 + 20} \times 0{,}714$$

$= 0{,}286\ A$ (3)

1.2.4 $R_T = R_1$

$= 2\ \Omega$✓ (1)

1.2.5 If R_5 is a short circuit (0 Ω), then R_2, R_3 and R_4 (parallel network) will be bypassed (no current will flow through these resistors)✓, so that current flows through R_1 and R_5 only. ✓ (2)

1.3 $P = S\cos\Theta$

$= 10\ 000 \times 0{,}85$

$= 8\ 500\ W$

$= 8{,}5\ kW$ (3)

1.4

1.4.1 $X_C = \frac{1}{2\pi fC}$

$= \frac{1}{2\pi \times 50 \times 30 \times 10^{-6}}$

= 106,10 Ω (2)

1.4.2 $X_L = 2\pi \times 50 \times 0{,}3H$

= 94,25 Ω (2)

1.5 An instrument transformer is used to reduce the value of current or voltage applied to a meter in known and definite ratios that bring the current and voltage to within the range of the meter.

OR

An instrument transformer is used to enable a small measuring instrument to measure high voltage and high current. (2)

1.6 1.6.1 The Wheatstone bridge is used to measure an unknown resistance. (1)

1.6.2
- It is used to monitor sensor devices such as strain gauges.
- It is used to measure temperature and temperature coefficient of resistance. (Any 1 × 1) (1)

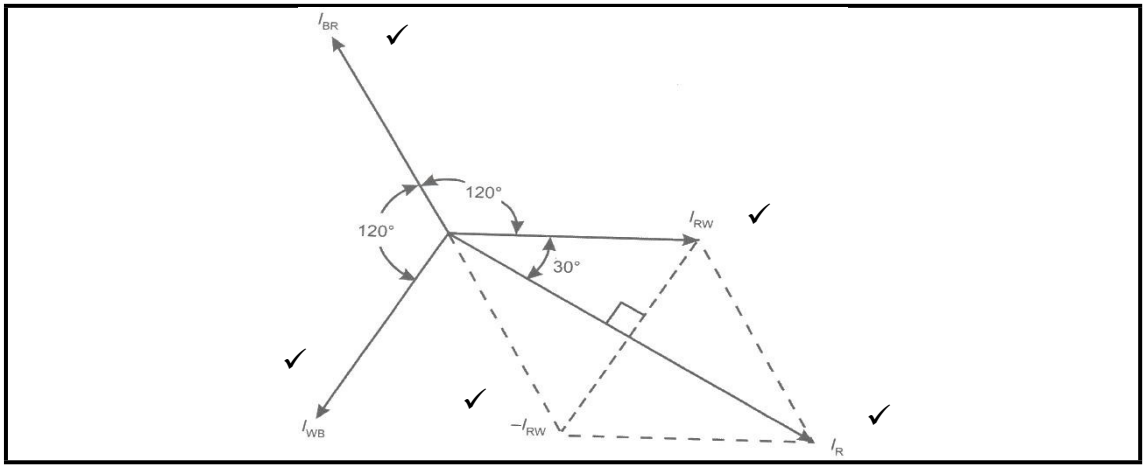
1.7 1.7.1 A frequency meter is connected in parallel in a circuit.
1.7.2 An ohmmeter is connected in parallel in a circuit.
(2 × 1) (2)

[30]

QUESTION 2

2.1 A – Three-phase generator
B – Step-up transformer
C – Secondary transmission system and primary distribution system
D – Step-down transformer
E – Consumer substation
F – District substation
G – Local distribution system (7 × 1) (7)

2.2



(5)

2.3 2.3.1 Lightning arrestor (1)

2.3.2 A lightning arrestor is used to provide a path for surges to be discharged to earth safely before a flashover on the line occurs. (2)

[15]

QUESTION 3

3.1
- To maintain the potential of any part of a system at a definite value with respect to earth
- To allow current to flow to earth in the event of a fault, so that the protective gear starts operating to isolate the faulty circuit
- To ensure that in the event of a fault, the apparatus cannot reach a dangerous potential with respect to earth
- To prevent electrocution of persons
- To prevent damage to equipment and apparatus (any 3x1) (3)

3.2
- Bare copper
- Galvanised iron or steel electrodes
- Stainless steel (3)

3.3 3.3.1 A consumer's earth terminal is a terminal at or near the point where the supply cables to the installation enter the building. (1)

3.3.2 - All conductive parts that must be earthed are connected to the main earthing terminal, which is connected to the consumer's earth terminal

- The consumer's earth terminal is earthed by connecting it to the supply earth terminal, or the protective conductor or the earth electrode. (2)

3.4
- Hot- and cold-water systems
- Antennas
- Roofs
- Gutters
- Downpipes
- Waste pipes
- Water pumps (Any 4 × 1) (4)

3.5 TN-C-S and TN-S (2)
[15]

QUESTION 4

4.1 Delta-delta
Star-star
Star-delta
Delta-star (4)

4.2 $Total\ \ losses = Iron\ \ losses + copper\ \ losses$

$= 1 + 1 = 2\ \ kW$

$P_{OUT} = VI\ \ cos\Theta$

$= 150 \times 0{,}87$

$= 130{,}5\ \ kW$

$P_{IN} = P_{OUT} + total\ \ losses$

$= 130{,}5 + 2$

$= 132{,}5\ \ kW$

$Efficiency\ \ (\eta) = \dfrac{P_{IN} - losses}{P_{IN}}$

or $\dfrac{P_{OUT}}{}$

$$= \frac{132,5 - 2}{132,5}$$

$$= 0,98 \text{ or } 98\ \%$$

P_{IN}

$=\underline{130,5}$ = 0,98 or 98%

132,5 (6)

4.3 4.3.1 $V_{P1} = V_{L1} = 3\ 300\ V$ (…..Delta connected)

$$V_{P2} = \frac{V_{L2}}{\sqrt{3}}$$

$$= \frac{380}{\sqrt{3}}$$

$$= 219,393\ V$$

$$\frac{N_1}{N_2} = \frac{V_{P1}}{V_{P2}}$$

$$= \frac{3\ 300}{219,393}$$

$$= 15,04 : 1$$ (5)

4.3.2 $$I_{L1} = \frac{S}{\sqrt{3}\ V_{L1}}$$

$$= \frac{400 \times 10^3}{\sqrt{3} \times 3300}$$

$$= 69,982\ A$$ (2)

4.3.3 $$I_{L2} = \frac{S}{\sqrt{3}\ V_{L2}}$$

$$= \frac{400 \times 10^3}{\sqrt{3} \times 380}$$

$$= 607,737\ A$$

ALTERNATIVE:

$$\frac{N_1}{N_2} = \frac{I_{P2}}{I_{P1}}$$

$$I_{P2} = \frac{15,04 \times 69,982}{\sqrt{3} \times 1}$$

$$I_{P2} = I_{l2} = 607,678\ A$$ (2)

4.3.4 $I_{P2} = I_{L2} = 607,737\ A$ ✓ (…..Star connected) (1)

[20]

QUESTION 5

5.1 5.1.1 False
5.1.2 True
5.1.3 True

(3 × 1) (3)

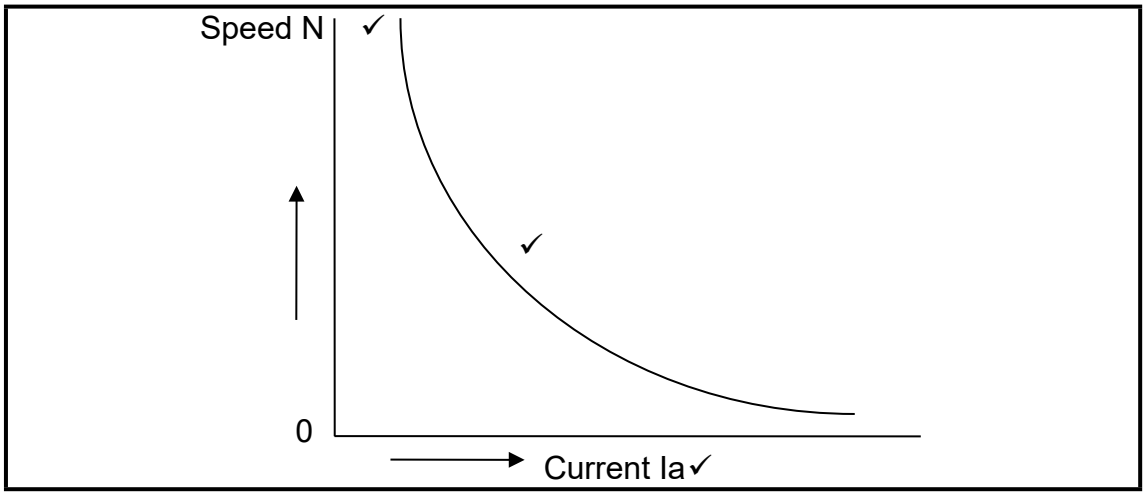
5.2
- A main winding or running winding
- An auxiliary winding or starting winding
- A squirrel-cage type rotor
- A centrifugal switch (4)

5.3 5.3.1 Low
5.3.2 Very high
5.3.3 High
5.3.4 Low
5.3.5 Not needed
5.3.6 Parts such as the slip rings and brush gear need maintenance

(6 × 1) (6)

5.4
- Magnetic or dashpot
- Thermal (2)

5.5



(3)

5.6 5.6.1 A generator is used to convert mechanical energy into electrical energy.

5.6.2 A motor is used to convert electrical energy into mechanical energy.

(2 × 1) (2)
[20]

TOTAL: 100

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